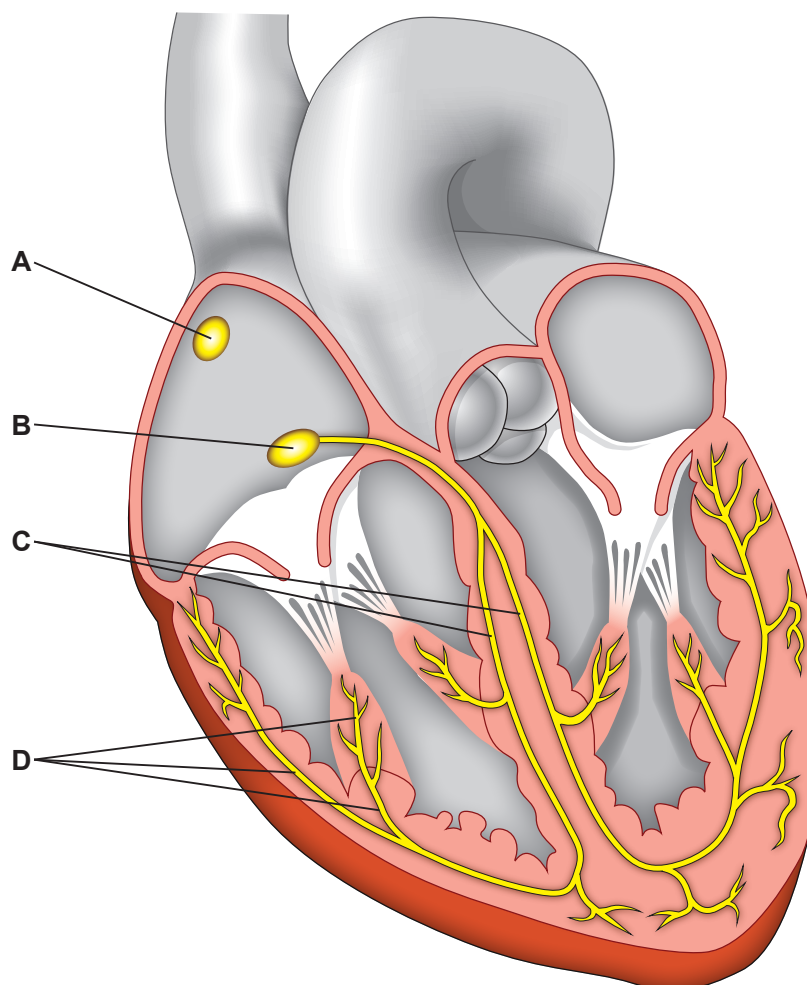


6. The cardiac cycle is controlled by several structures within the heart. **Image 6.1** shows a longitudinal section through the heart.

Image 6.1



- (a) (i) Cardiac muscle has the property of being myogenic.
State what is meant by the term myogenic.

[1]

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- (ii) Use letters (**A–D**) from **Image 6.1** to identify the following structures:

[2]

- I. the atrio-ventricular node
- II. the Purkyne fibres
- III. the sino-atrial node



- (iii) Describe and explain the role of structures labelled **A–D** in **Image 6.1** in controlling the cardiac cycle.

[4]

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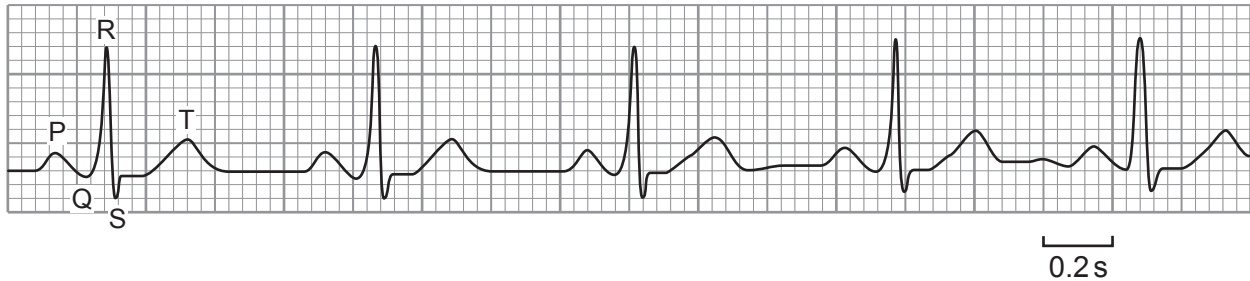
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- (b) **Image 6.2** shows a normal ECG (electrocardiogram) trace.

Image 6.2



Use **four** cardiac cycles shown on **Image 6.2** to calculate the heart rate of this individual.

[2]

Heart rate = beats per minute

- (c) A heart block is a medical condition which is caused by an obstruction in the electrical conduction system of the heart. These obstructions can either affect the sino-atrial node or the atrio-ventricular node.

Image 6.3 shows an abnormal ECG trace. This was taken from a patient suffering from a heart block.

Image 6.3



- (i) **Image 6.3** shows that there is an obstruction at the atrio-ventricular node and not at the sino-atrial node. Explain the evidence for this.

[2]

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(ii) Suggest the effect that this obstruction would have on the cardiac cycle.

[2]

Examiner
only

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